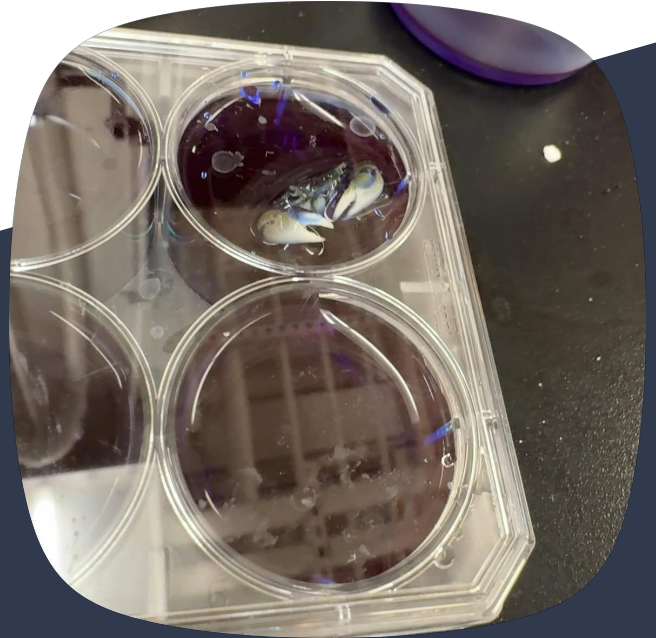


Pollution and its Impact on Predator Cue Detection

By Noah Ozguner, Abigail Kwan, Callum Rangan, Jack Brown



The Hairy Shore Crab (*Hemigrapsus oregonensis*)

- Native to the Puget Sound, ranging from Alaska to Baja California
- Found within the intertidal, specifically in mud flats, eelgrass beds, and open beaches
- Often feeds on green algae and diatoms while being preyed on by larger crabs, fish, and seabirds
- Will use a hair-like appendage known as a Setae and antennae for chemoreception to find food or avoid predators



The European Green Crab (*Carcinus maenas*) Invasion

Highly resilient to changes in temperature and salinity

Reproduce rapidly allowing for a large invasion in just a few years

- Populations were estimated to be constantly on the rise despite elimination efforts



Cause incredible harm on local ecosystems:

- Destabilizing banks by burrowing
- Eat Dungeness, other native crabs, and native shellfish
 - Dramatically reduces populations leading to a decline in key crab and shellfish commercial industries
- Digs up eelgrass beds
 - Reduces habitat for native fish, seabird, and other invertebrates

The Rising Threat of Pollution on Chemoreception

Chemoreception is used by a hairy shore crab's Setae to detect chemical cues found in the water

Cues can be used to detect a range of things:

- Food
- Predators
 - through chemical cues known as predator cues
- Environmental Changes
 - Temperature
 - Salinity

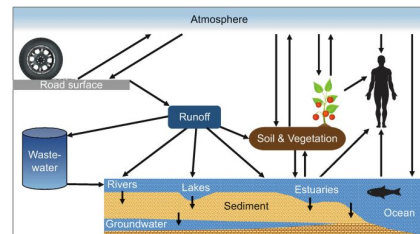
Under higher levels of pollution, these chemical cues can be masked by varying chemicals

The masking of predator cues may inhibit the hairy shore crab's ability to detect the European green crabs to avoid predation



Organic Pollutants

Puget sound pollutants	Sources	Pathways	Sample Used
Copper	- Anti-fouling paints - Brake-pads	- Runoff - Boat transit	3x3 Aluminium with 2 coats of anti-fouling paint
Tire chemicals	6PPD and 6PPDq - Tire and road wear particles (TWPs)	- Runoff - Atmospheric deposition	6.849 g of tire pieces
Herbicides	2,4-D, Atrazine - Agricultural & Residential use	- Runoff - Groundwater filtration	- High: 500 ug/L - Low: 150 ug/L
Motor Oil	- Vehicle leaks - Improper disposal - Industrial spills	Runoff	- High: 500 ug/L - Low: 150 ug/L



Research Question, Hypothesis, and Prediction

Research Question

How do Puget Sound pollutants affect *Hemigrapsus* spp.'s ability to sense predators, through chemosensing?

Hypothesis

H_0 = There will be no change in the metabolic rate or antennae flicking per minute based on the pollution when *H. spp* in the presence of *C. maenas*

H_a = There will be a change in the metabolic rate or antennae flicking per minute based on the pollution when *H. spp* in the presence of *C. maenas*

Prediction

There will be a **decrease** in the metabolic rate and antennae flicking per minute when *H. spp* in the presence of *C. maenas* while exposed to pollutants

Experimental Design

2x2 Factorial design
x2 *H. Oregonensis* and 1 *H. Nudus* in each tank 🦀🦀
x1 *C. maenas* per tank for tanks 1B & 2B 🦀

1 = No Pollutants

2 = Pollutants

A = No Predator

B = Predator

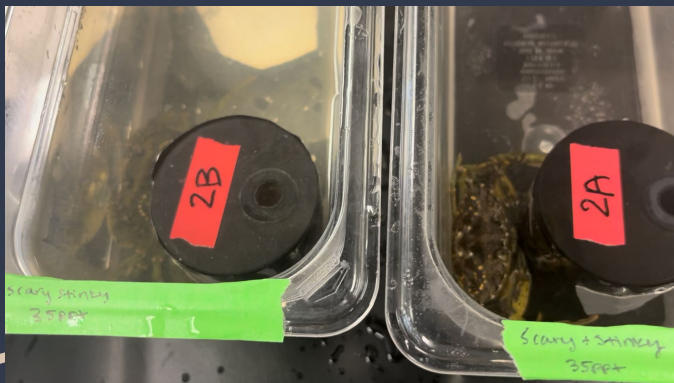


Data to Collect from *H. oregonensis*

- Metabolic Rate
- Antenna Flicking Rate
- Mass · Sex

Zhang, et al. (2021)

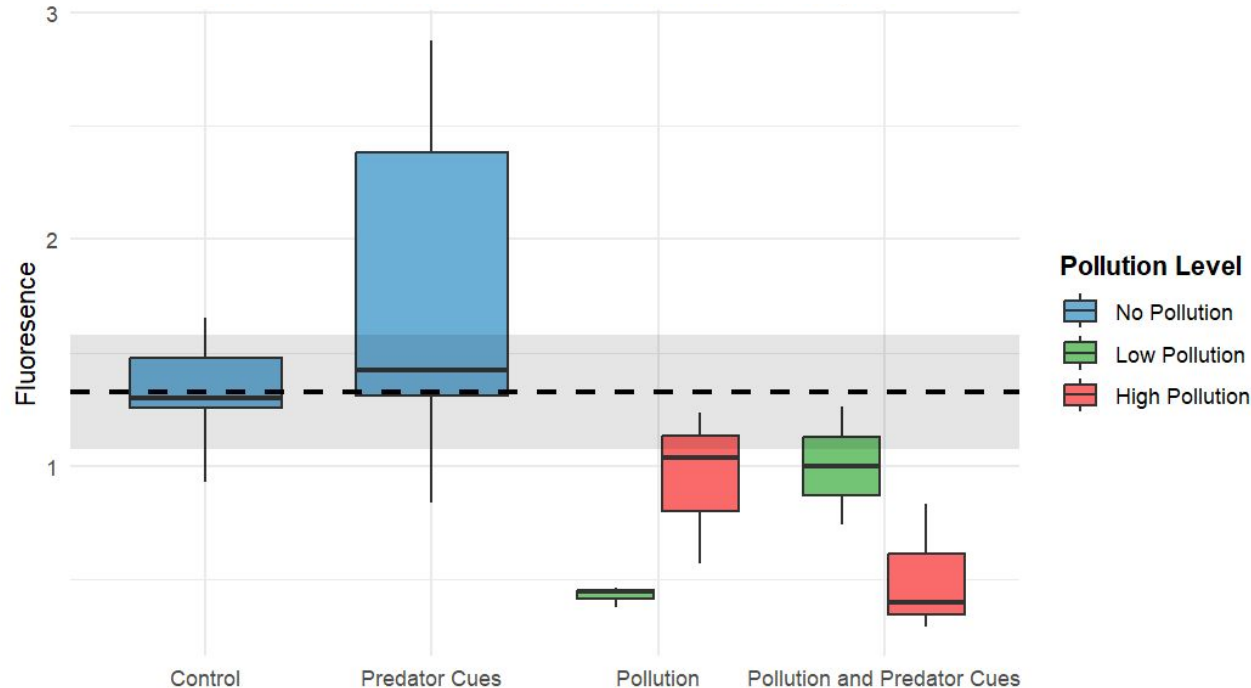
Scott, K., Harsanyi, P., & Lyndon, A. R. (2018)



Data Collection

1. Crabs were first assayed for antenna flicks per minute using video recordings
 - *C. maenas* were placed in tanks before crabs were assayed for antenna flicks
2. Resazurin assay was then use to determine metabolic rate
 - Shore crabs were placed in the 6-well plate containing ~12 mL of resazurin working solution
 - 200 uL sample was transferred from each chamber into a 96-well plate
 - Samples were collected every 30 minutes at three time points: T0, T1, T2

Metabolic Rate per Treatment Group

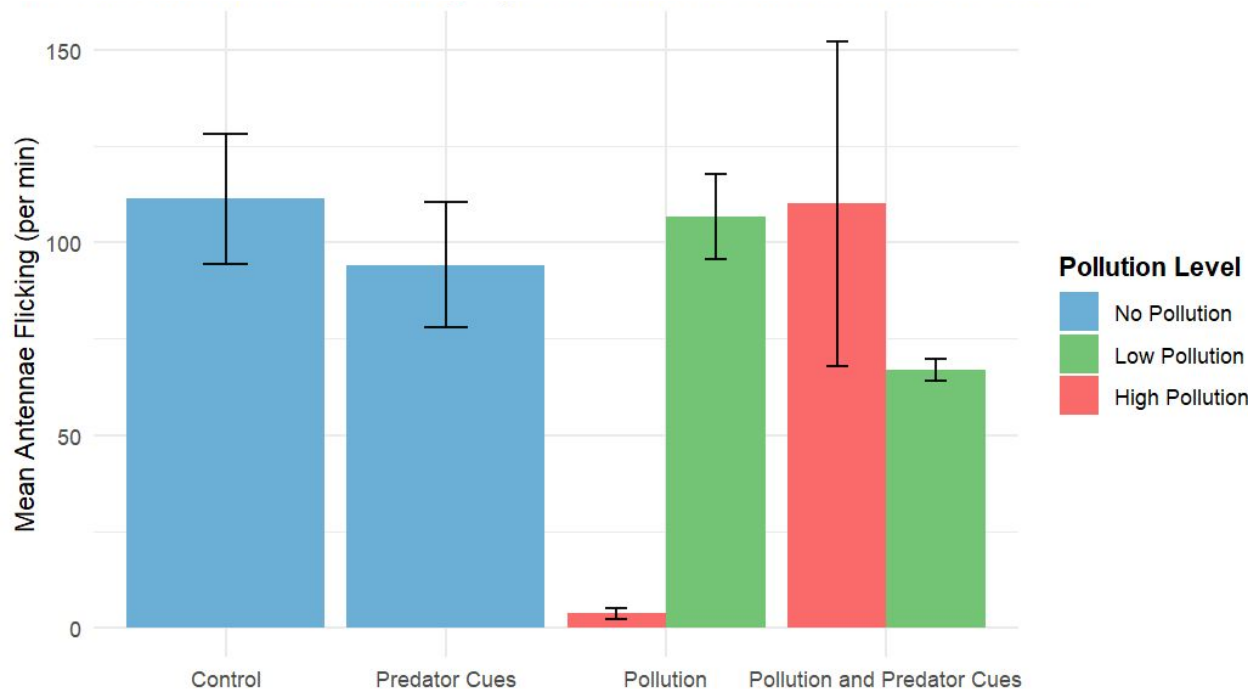


Metabolic rate visually lower in pollution treatments and varied with predator cues

ANOVA and Tukey tests indicate that the **predator cue group's metabolic rate is significantly higher than high pollution with predator cues group** ($p = 0.026$)

No other comparisons indicated any significance, including comparisons to the control

Mean Antennae Flicking per Treatment Group



High pollution and low pollution with predator cues were the only ones notably lower antennae flicking

ANOVA and Tukey tests indicates that the antennae flicking in the **high pollution with predator cues is significantly higher than the high pollution group** ($p = 5.4e-5$)

No other comparisons indicated any significance or relevance, including comparisons to the control

Conclusions & Future work

Pollution may affect predator avoidance in hairy shore crabs

- Crabs exposed to both predators and pollution exhibited lower metabolic rates in high pollution
 - Less possible energy to respond to a predator
- Antennae flicking rates in pollutant treatments were comparable to the control group, regardless of treatment
 - Shore crabs may have still been able to detect the presence of *Carcinus maenas* nearby
- Crabs exposed to high pollutant levels had the lowest flicking rate of any treatment
 - Shore crabs refused to “smell” surrounding water
 - 2/3 of the crabs in this group had not been acclimated to the polluted water due to mortality replacements

Pollution Exposure Levels and Time

- Environmentally accurate pollution levels instead of pushing to the extreme
- Increase amount of time exposed to pollutants to understand any long term effects

Improve Study Accuracy

- Further and more detailed testing
- Dramatically increase sample size as well as standardization

Ecological Concerns

- As European green crab populations grow, shore crab populations should continue to be monitored as the pollution present may impede their ability to resist.